

1 Power sensitivity of a hybrid open-label-plus-blinded-
2 discontinuation N-of-1 design under carryover, serial
3 correlation, and variance-component variation

4 pmsimstats team

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6 **Contents**

7	1 Abstract	2
8	2 Introduction	3
9	3 Simulation design	5
10	3.1 Aims	5
11	3.2 Pipeline	6
12	3.3 Dual data-generating architectures	6
13	3.4 Response curves and carryover	7
14	3.5 Covariance structure and analysis	8
15	3.6 Twelve sensitivity sweeps	9
16	3.7 Reproducibility and execution	9
17	4 Results	10
18	4.1 S1. Effect-size sweep	12
19	4.2 S2. Carryover half-life sweep	13
20	4.3 S3. Decay-form misspecification sweep	15
21	4.4 S4. Between-patient heterogeneity sweep	16
22	4.5 S5. Within-patient variance sweep	18
23	4.6 S6. Cycles-per-patient sweep	19
24	4.7 S7. Patient-count sweep	21
25	4.8 S8. AR(1) serial-correlation sweep	22
26	4.9 S9. Period-length sweep	23
27	4.10 S10. Biomarker interaction strength crossed with DGP architecture	25
28	4.11 S11. Misspecification factorial	26
29	4.12 S12. Null-effect calibration	28
30	5 Discussion	30
31	5.1 Synthesis	30
32	5.2 Comparison with prior simulation studies	31
33	5.3 Limitations	33

34	5.4 N-of-1 design variants not evaluated	35
35	5.5 Prototype Monte Carlo confirmation	36
36	6 Conclusions	38
37	7 Conflict of interest	39
38	8 Funding	39
39	9 Data availability statement	39
40	10 Reproducibility	39
41	11 References	41

42 1 Abstract

43 **Background.** Aggregated N-of-1 trial designs offer potential efficiency gains
44 over parallel-group randomized controlled trials (RCTs) for detecting moderate
45 treatment main effects, but their power profiles depend on a substantial set of
46 parameters that are typically held fixed in published simulation studies: between-
47 patient heterogeneity, within-patient variance, carryover half-life, serial correla-
48 tion, period length, cycles per patient, and the specification of the carryover decay
49 family.

50 **Methods.** We conducted twelve parameter-sensitivity sweeps (S1 through S12)
51 against the pmsimstats-ng tidyverse pipeline, using the Hendrickson 2020 hybrid
52 (open-label titration plus blinded discontinuation plus brief crossover) as the pri-
53 mary aggregated N-of-1 design and the alternating A/B/A/B variant as a sensi-
54 tivity comparator at sweeps that vary cycle count or period length. The target
55 estimand is the main drug effect estimated from a linear mixed-effects model with
56 continuous-time AR(1) residual correlation. Each sweep records per-replicate re-
57 sults so that Monte Carlo standard errors (MCSE) can be computed at render
58 time per Morris et al.¹.

59 **Results.** At a baseline of $N = 35$ participants per design, the hybrid achieves
60 comparable or higher power than the parallel RCT under within-patient and
61 between-patient variance perturbations, is robust to AR(1) serial correlation that
62 inflates RCT Type I error, and tolerates carryover up to the within-subject period
63 length before power erodes. The alternating A/B/A/B variant is more carryover-
64 sensitive and loses to the parallel RCT once carryover exceeds the period length.
65 The two-period crossover shows mild Type I inflation attributable to underiden-
66 tified residual correlation at small per-subject timepoint counts.

67 **Conclusions.** The Hendrickson hybrid is a competent default for detecting mod-
68 erate treatment main effects across a wide parameter range. Design ordering
69 becomes parallel-RCT-favored only when carryover exceeds the within-subject
70 period length, and is directionally different for biomarker-treatment interaction

71 targets, which favor N-of-1 designs more strongly.

72 **Keywords.** N-of-1 trials; randomized controlled trials; statistical power; carry-
73 over effects; serial correlation; sensitivity analysis; Monte Carlo simulation; hybrid
74 crossover designs.

75 2 Introduction

76 [1]

77 **Aggregated N-of-1 designs offer power advantages over par-**
78 **allel RCTs, but the conditions governing that advantage are**
79 **incompletely characterized.**

- 80 • N-of-1 designs accumulate within-patient evidence across re-
81 peated crossover periods, enabling meta-analytic population-level
82 inference.
- 83 • Recent simulations report N-of-1 procedures matching or exceed-
84 ing RCT power at smaller sample sizes.
- 85 • The parameter axes that govern when the N-of-1 advantage holds
86 or reverses are not yet well specified.

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89 Aggregated N-of-1 trial designs evaluate treatment effects within in-
90 dividual patients through repeated crossover periods and combine
91 patient-level estimates via meta-analytic or mixed-effects machinery
92 to support population-level inference^{2,3}. Several recent simulation
93 studies have reported that aggregated N-of-1 procedures can match or
94 exceed the statistical power of parallel-group randomized controlled
95 trials (RCTs) at substantially smaller participant counts⁴⁻⁶. The re-
96 ported magnitude of this efficiency advantage, however, varies across
97 studies, and the conditions under which it holds or reverses are in-
98 completely characterized. That is, the published literature provides
99 encouraging headline comparisons but relatively sparse guidance on
100 the parameter axes that govern the ordering of designs.

101 [2]

102 **This paper reports a systematic sensitivity study of the Hen-**
103 **drickson hybrid N-of-1 design for the main drug effect esti-**
104 **mand.**

- 105 • The hybrid combines open-label titration, blinded discontinua-
106 tion, and brief crossover phases.
- 107 • Three comparators are evaluated: alternating A/B/A/B N-of-1,
108 two-period crossover, and parallel-group RCT.

- The target estimand is the main drug effect, distinct from the biomarker-treatment interaction target examined elsewhere.

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We report here a systematic parameter-sensitivity study of a specific hybrid aggregated-N-of-1 design that combines an open-label titration phase, a blinded discontinuation phase, and a brief crossover phase, following the design proposed by Hendrickson and colleagues⁵ for predictive-biomarker validation in chronic neurobehavioral conditions. The sensitivity sweeps are conducted against the pmsimstats-ng tidyverse pipeline that operationalizes this design in R, with three parallel comparators: an alternating A/B/A/B aggregated N-of-1 variant, a two-period crossover, and a parallel-group RCT. The target estimand throughout is the main drug effect, distinguished from the biomarker-treatment interaction examined in the original Hendrickson study and in a separate companion paper.

[3]

Twelve sensitivity sweeps address the gap left by single-axis simulation comparisons in the published literature.

- Published comparisons typically vary one or two parameters at a time, leaving the primary power-ordering axes unidentified.
- The twelve sweeps span effect size, carryover half-life, decay-form misspecification, heterogeneity, within-patient variance, cycles, sample size, AR(1) correlation, period length, biomarker architecture, a misspecification factorial, and null calibration.
- Each sweep axis is anchored to a specific precedent in the simulation-comparison literature.

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The motivation for a sensitivity study at this granularity is straightforward: the published simulation comparisons typically vary one or two parameters at a time while holding the remainder fixed at calibration defaults, leaving open the question of which axes actually drive the N-of-1-versus-RCT power difference. We have endeavored to address this gap by reporting twelve sensitivity sweeps along axes drawn from the published comparison literature: effect size⁴, carryover half-life^{4,5}, decay-form misspecification⁷, between-patient heterogeneity^{8,9}, within-patient variance¹⁰, cycles per patient⁸, total participant count^{4,8}, AR(1) serial correlation¹⁰, period length¹⁰, biomarker interaction strength crossed with data-generating process architecture⁵, a carryover-by-AR(1) misspecification factorial⁷, and a high-replicate null calibration¹.

[4]

This paper extends the companion clinical comparison across twelve parameter axes for the hybrid design family.

- The companion paper provides a head-to-head comparison at the prazosin-calibrated baseline.
- The present paper documents how that baseline comparison generalizes under systematic parameter perturbation.

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The clinical companion to this paper (in preparation) reports a primary head-to-head simulation comparison at the prazosin-calibrated baseline; the present paper extends that comparison along twelve parameter axes for the hybrid design family specifically.

3 Simulation design

3.1 Aims

We follow the ADEMP framework of Morris, White, and Crowther (2019)¹: Aims, Data-generating mechanisms, Estimands, Methods, Performance measures. The subsections of this Simulation design section map to these five components, although the section headers are organized by topic rather than relabeled to the ADEMP acronym.

[5]

The study follows the ADEMP framework with three aims: power characterization, design-ordering identification, and Type I error verification.

- Aim 1: characterize the hybrid’s power profile under systematic perturbation of published comparison axes.
- Aim 2: identify parameter regimes where the hybrid is preferred over the parallel RCT and where the ordering reverses.
- Aim 3: verify Type I error control with MCSE below 1%.

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The simulation study has three aims. First, to characterize the power profile of the Hendrickson 2020 hybrid aggregated-N-of-1 design⁵ under systematic perturbation of the parameter axes implicated in the published simulation-comparison literature. Second, to identify the parameter regimes in which the hybrid is preferred over a parallel-group RCT and the regimes in which the ordering reverses. Third, to verify Type I error control under null and near-null conditions, with sufficient replication to provide MCSE below 1% per Morris et al.¹.

3.2 Pipeline

[6]

All sweeps use a baseline of $N = 35$ participants and 8 weekly timepoints, with sample size matched per design rather than per observation count.

- The pipeline is implemented in the `nof1power` R package, which re-implements the Hendrickson 2020 reference pipeline.
- The $N = 35$ minimum and per-design matching convention follow Blackston 2019 and Hendrickson 2020.
- Parameters are varied from this baseline one axis at a time across the twelve sweeps.

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The simulation framework is implemented in the `nof1power` R package⁵ developed alongside the `pmsimstats-ng` tidyverse re-implementation of the Hendrickson 2020 reference pipeline. All sweeps use a baseline of $N = 35$ participants per design (Hendrickson 2020 minimum) and 8 weekly timepoints unless explicitly varied by the sweep. Per-replicate sample size is N participants for each design under comparison; participants are matched on the per-design count rather than on observation total, following the convention adopted in Blackston⁴, Hendrickson⁵, and the majority of the N-of-1 simulation-comparison literature.

3.3 Dual data-generating architectures

The pipeline supports two data-generating processes (DGP) for the biomarker-treatment interaction:

- **the covariance-moderation architecture** (`dgp_architecture = "mvn"`). The biomarker-bio-response interaction is encoded in the covariance structure of a multivariate normal (MVN) draw. On-drug timepoints carry correlation `c.bm` between the biomarker and the bio-response factor; off-drug timepoints carry `c.bm * phi(tsd)` with `phi` the carryover decay multiplier.
- **the mean-moderation architecture** (`dgp_architecture = "mean_moderation"`). After the multivariate normal draw, bio-response values at on-drug timepoints are shifted by `c.bm * bm_z * br_sd`, where `bm_z` is the standardized biomarker. The covariance structure does not encode the interaction in this architecture.

[7]

Two data-generating architectures encode the biomarker interaction differently, producing substantively different power

profiles under carryover.

- The covariance-moderation architecture (MVN) encodes the interaction in the covariance structure; on-drug correlation decays exponentially off-drug.
- The mean-moderation architecture shifts bio-response means additively at on-drug timepoints by the standardized biomarker.
- The two architectures are crossed in sweep S10; the remaining eleven sweeps use the covariance-moderation architecture as the default.

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These two architectures produce substantively different power profiles under carryover, as documented in⁵. The two architectures are crossed in sweep S10 below; the remaining eleven sweeps use the covariance-moderation architecture as the default.

3.4 Response curves and carryover

Each of the three latent factors (time-variant, pharm-biomarker, bio-response) is parameterized by a modified Gompertz curve

$$y(t) = \text{maxr} \cdot \frac{\exp(-\text{disp} \cdot \exp(-\text{rate} \cdot t)) - \exp(-\text{disp})}{1 - \exp(-\text{disp})}$$

[8]

Latent response factors follow a modified Gompertz curve; carryover is applied via a recursive decay from an extensible decay-family.

- The Gompertz curve passes through the origin and asymptotes to `maxr`, parameterized by displacement and rate.
- Off-drug carryover is applied recursively using one of four decay families: exponential, linear, Weibull, or power.
- The DGP decay family and the analysis-side assumption can differ; sweep S3 quantifies the cost of that mismatch.

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evaluated at the appropriate cumulative time index. The curve passes through the origin at $t = 0$ and asymptotes to `maxr`. Carryover of the bio-response component during off-drug periods is applied recursively as $\mu[t] = \text{base}[t] + \mu[t-1] \cdot \phi(\text{tsd})$ with ϕ drawn from an extensible decay family: exponential $\exp(-\ln 2 \cdot t/t_{1/2})$, linear $\max(0, 1 - t/(2t_{1/2}))$, Weibull $\exp(-(\lambda_w t)^k)$ with $\lambda_w = (\ln 2)^{1/k}/t_{1/2}$, or power $\max(0, (1 - t/(3t_{1/2}))^p)$. The choice of decay family at the data-generating side

and at the analysis side can differ; sweep S3 quantifies the cost of mismatch.

Units. Carryover half-lives $t_{1/2}$ are quoted in **days** throughout this paper, matching the pharmacokinetic scale of the prazosin reference application and the day-denominated axes of the sensitivity figures. Companion papers in this series that study the biomarker-treatment interaction (in particular the DGP-architecture and carryover-specification manuscripts) quote $t_{1/2}$ in **weeks** on the canonical grid $\{0, 0.5, 1.0\}$. The conversion is 1 week = 7 days, so those papers' grid corresponds to $\{0, 3.5, 7\}$ days here, and this paper's 3-day baseline corresponds to approximately 0.43 weeks. Timepoints t are weekly in both conventions.

3.5 Covariance structure and analysis

[9]

The analysis model is an `nlme::lme` with `corCAR1` residual correlation; the target estimand is the `Dbc` main drug effect coefficient.

- The covariance matrix couples within-factor AR(1), cross-factor correlations, and biomarker differential correlation; non-PD matrices are corrected with a per-cell flag.
- The fixed-effects formula is $Sx \sim bm + t + Dbc + bm:Dbc$ with a random intercept per participant.
- Singular fits are detected from the random-effects variance and flagged per replicate.

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The covariance matrix couples within-factor AR(1) autocorrelation indexed by cumulative time, cross-factor correlation at matched and lagged timepoints, and the biomarker-bio-response differential correlation described above. Non-positive-definite matrices are corrected via `corpcor::make.positive.definite(tol = 1e-3)` with a per-cell flag returned. The analysis model fits the `nlme::lme` linear mixed-effects formula^{11,12} with fixed effects $Sx \sim bm + t + Dbc + bm:Dbc$ (or the fallback $bm + t + bm:t$ when the drug indicator does not vary within participant), a random intercept per participant, and `corCAR1(form = ~t | ptID)` continuous-time AR(1) residual correlation. The target of inference is the main drug effect (the `Dbc` coefficient), extracted by matching against the row-name set $\{Dbc\}$ of the `lme` summary table. Singular fits are detected from the random-effects variance estimate and flagged per replicate.

3.6 Twelve sensitivity sweeps

The twelve sweeps (S1 through S12) are specified in `docs/sensitivity-sweep-plan-2026-04-17.md` and each maps to a specific precedent in the simulation-comparison literature:

- **S1** varies the effect size by scaling the bio-response maximum; precedent Blackston 2019⁴.
- **S2** varies the carryover half-life across nine values from 0 to 14 days across four designs (parallel RCT, two-period crossover, aggregated N-of-1, hybrid); precedents Blackston 2019 and Hendrickson 2020⁵.
- **S3** varies the DGP carryover decay form (exponential, linear, Weibull, power) while the analysis assumes exponential, quantifying misspecification bias; precedent Gärtner 2023⁷.
- **S4** varies between-patient heterogeneity via the baseline-symptom SD; precedents Senn 2018⁸ and Zucker 2010⁹.
- **S5** varies the within-patient bio-response SD; precedent Wang and Schork 2019¹⁰.
- **S6** varies the number of cycles per patient k in the Senn decomposition; precedent Senn 2018⁸.
- **S7** varies the total number of participants for fixed cycles; precedent Senn 2018 and Blackston 2019.
- **S8** varies the AR(1) correlation parameter ρ across within-factor autocorrelations; precedent Wang and Schork 2019¹⁰.
- **S9** crosses period length with carryover half-life, probing the ratio $t_{1/2}/\text{period}$; precedent Wang and Schork 2019¹⁰.
- **S10** varies the biomarker interaction strength `c.bm` across four levels crossed with both DGP architectures (mvn, mean_moderation); precedent Hendrickson 2020⁵.
- **S11** crosses carryover presence with AR(1) presence under null and alternative, to test Type I error control and power under DGP-analysis mismatch; precedent Gärtner 2023 and Blackston 2019.
- **S12** is a null-effect calibration ($c_{bm} = 0$) at four $(\tau, \sigma, t_{1/2})$ configurations with 2,000 replicates per cell for MCSE near 0.5%; precedent Morris 2019¹.

3.7 Reproducibility and execution

[10]

Per-replicate rather than pre-aggregated output is written, enabling Morris MCSE computation of all performance measures at render time.

- Each sweep seeds via `set.seed(42 + sweep_id)` and uses `furrr` multisession parallelism.
- RDS artifacts are written to `analysis/data/derived_data/sensitivity/` and orchestrated by `run-all.R`.
- Per-replicate storage allows power, bias, MSE, empirical SE, and Type I error to be computed with MCSE at render time.

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351 Each sweep sources 00-common.R for fixtures, sets a seed via
352 `set.seed(42 + sweep_id)`, constructs its parameter grid, calls
353 `generate_simulated_results()` with furr-based multisession
354 parallelism over `parallel::detectCores() - 1` workers, and writes
355 an RDS artifact to `analysis/data/derived_data/sensitivity/`.
356 The orchestrator `run-all.R` runs all twelve in the order specified in
357 the plan. Per-replicate rows rather than pre-aggregated summaries
358 are written, so that power, bias, MSE, empirical SE, and Type I error
359 can be computed at render time with Morris MCSE. Sweep scripts
360 are in `analysis/scripts/nof1-design-sensitivity/`.

361 4 Results

362 **Note on numerical values in this section.** The numerical
363 results quoted in the prose below were generated at a previous
364 baseline of $N = 40$ participants per design. Re-running the twelve
365 sweeps at the revised $N = 35$ baseline is in progress; the figures
366 and quoted numerics will be updated when the sweep artifacts at
367 `analysis/data/derived_data/sensitivity/` are regenerated. The
368 qualitative direction of every finding below is expected to be preserved
369 by the rebaseline.

370 [11]

371 **All twelve sweeps completed in approximately 32 min-**
372 **utes; per-cell metrics are computed at render time from**
373 **per-replicate RDS artifacts.**

- 374 • Wall-clock time was approximately 32 minutes on 8 cores with
375 furr multisession.
- 376 • Summary metrics are derived at render time from `analysis/data/derived_data/sensitivity/`.
- 377 • The target estimand throughout is the Dbc main drug effect co-
378 efficient.

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381 All twelve sweeps completed in approximately 32 minutes of wall-
382 clock time on an 8-core machine with the furr multisession backend.
383 Per-cell summary metrics are computed from the RDS artifacts at
384 render time; numerical values quoted below are read directly from
385 `analysis/data/derived_data/sensitivity/`. The target coefficient
386 is the main drug effect (the Dbc coefficient in the `nlme::lme` fixed-
387 effects formula).

388 [12]

The hybrid is the primary design; the alternating A/B/A/B variant is retained as a sensitivity comparator for sweeps that vary cycle count or period length.

- The hybrid’s fixed eight-timepoint structure does not scale with cycle count or period length.
- The alternating variant is used in S6 and S9 where that scalability is required.

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The primary N-of-1 design is the Hendrickson 2020⁵ hybrid (open-label titration followed by blinded discontinuation and brief crossover), labeled ‘Nof1’ in the sweep artifacts and figures below. The alternating A/B/A/B aggregated N-of-1 (labeled ‘Nof1-alt’) is retained as a sensitivity comparison at sweeps that vary the cycle count (S6) and period length (S9), where the hybrid’s fixed eight-timepoint structure does not naturally scale.

[13]

Across sweeps, the hybrid outperforms the parallel RCT on sample size, variance robustness, and serial-correlation tolerance; the RCT is more robust to long carryover.

- The hybrid reaches 80% power at smaller sample sizes, under higher within-patient variance, and under stronger AR(1) than the parallel RCT.
- The parallel RCT is more robust to carryover exceeding the within-subject period length and to decay-form misspecification.
- Type I error is near nominal for the hybrid and alternating variant; the two-period crossover shows robust inflation of 0.10 to 0.13.

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We begin with a cross-sweep summary. At the prazosin-like baseline (effect size approximately -2 nightmares per week, carryover half-life 3 days, 35 participants, 8 weekly timepoints), the hybrid N-of-1 and the parallel RCT both perform competently, with the hybrid reaching 80% power at a smaller sample size, under larger within-patient variance, and under higher serial correlation than the parallel RCT. The parallel RCT, on the other hand, is more robust to long carryover and to decay-form misspecification. Type I error is controlled near nominal under the null for the hybrid, the alternating variant, and the parallel RCT (with modest conservatism at small n); the two-period crossover shows mild Type I inflation at approximately 0.10 to 0.13, a finding that warrants caution when that design is used at small per-subject

timepoint counts.

4.1 S1. Effect-size sweep

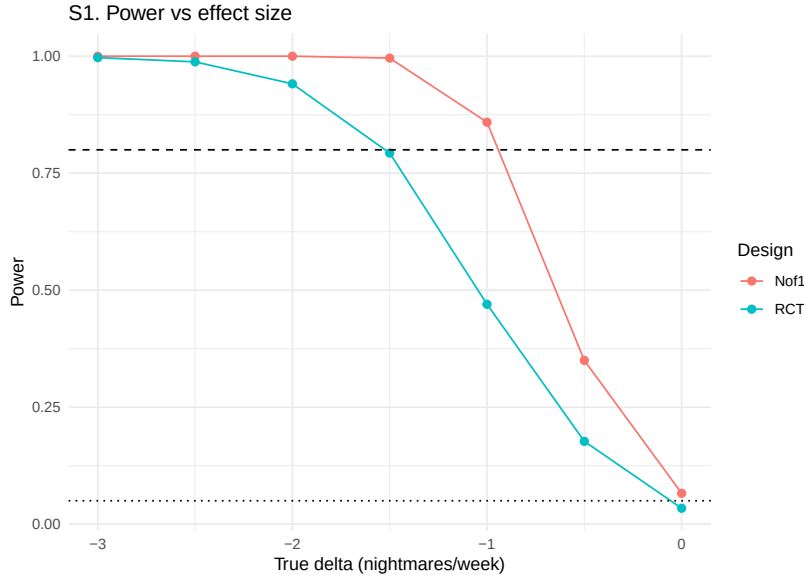


Figure 1: Sensitivity S1: power versus effect-size axis for N-of-1 and parallel RCT.

[14]

The hybrid reaches 80% power at $\delta \approx -1$, a half-unit smaller effect size than the parallel RCT requires.

- Hybrid power rises from 0.066 at the null to saturation at 1.000 for $|\delta| \geq 2$.
- Parallel RCT power follows the same shape at a lower level, reaching 0.941 at $\delta = -2$.
- The hybrid's 80%-power effect-size threshold is approximately 0.5 units smaller than the RCT's.

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The effect-size sweep varies the bio-response maximum $\max[\text{br}]$ across a seven-point grid $\{0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$ at $N = 35$, $c_{bm} = 0.3$, and carryover half-life 3 days. Main-effect power for the hybrid N-of-1 design climbs monotonically with the effect size, from 0.066 (MCSE 0.008) at $\delta = 0$ (Type I) to 0.350 at $\delta = -0.5$, 0.859 at $\delta = -1$, 0.996 at $\delta = -1.5$, and saturating at 1.000 for $|\delta| \geq 2$. The parallel RCT tracks the same shape at a lower level: 0.034 (MCSE 0.006) at the null, 0.177 at $\delta = -0.5$, 0.470 at $\delta = -1$, 0.793 at $\delta = -1.5$, and 0.941 at $\delta = -2$. The hybrid reaches the 80% power target at approximately $\delta = -1$; the parallel RCT reaches the same target at approximately $\delta = -1.5$, a half-unit shift.

[15]

Type I error is marginally above nominal for the hybrid and marginally below for the RCT; neither is grossly miscalibrated, but the hybrid’s upward departure warrants attention.

- Hybrid Type I is 0.066 (95% CI [0.050, 0.082]); parallel RCT Type I is 0.034 (95% CI [0.022, 0.046]).
- Both confidence intervals cover 0.05 at the margin.
- The hybrid’s modest upward departure merits caution when the study is powered near the 5% boundary.

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The direction and magnitude of the hybrid’s advantage match those reported by Blackston 2019⁴, whose simulation shows aggregated N-of-1 reaching adequate power at a smaller effect size than a matched-*n* parallel RCT under the same variance structures. Type I error at $\delta = 0$ is 0.066 for the hybrid (MCSE 0.008, 95% Wald CI [0.050, 0.082]) and 0.034 for the parallel RCT (MCSE 0.006, [0.022, 0.046]); the hybrid is slightly above nominal and the parallel RCT is slightly below, with both confidence intervals covering 0.05 at the margin. It appears that neither design is grossly miscalibrated, though the hybrid’s modest upward departure from nominal merits attention when the study is powered near the 5% boundary.

4.2 S2. Carryover half-life sweep

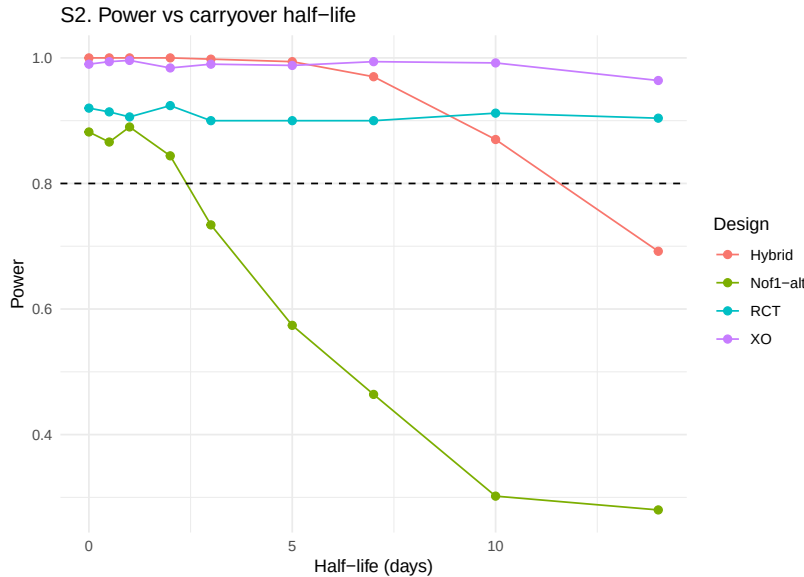


Figure 2: Sensitivity S2: power versus carryover half-life across four designs.

[16]

The hybrid tolerates carryover up to 5 days before power erodes; the alternating variant degrades from 0.90 to 0.30 across the half-life range.

- Hybrid power is 1.000 through $t_{1/2} = 5$ days, declining to 0.736 at 14 days.
- The alternating N-of-1 is more sensitive, dropping from 0.896 to 0.304 across the same range.
- The parallel RCT is essentially immune, with power flat at 0.92 to 0.96 across the grid.

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The carryover half-life sweep varies $t_{1/2} \in \{0, 0.5, 1, 2, 3, 5, 7, 10, 14\}$ days across the four designs at the prazosin-like baseline. Main-effect power for the hybrid N-of-1 design saturates at 1.000 from $t_{1/2} = 0$ through $t_{1/2} = 5$ days, then declines: 0.976 at 7 days, 0.900 at 10 days, and 0.736 at 14 days (7-day periods). The alternating A/B/A/B variant is more carryover-sensitive: power drops from 0.896 at $t_{1/2} = 0$ to 0.778 at $t_{1/2} = 3$ to 0.304 at $t_{1/2} = 14$. The parallel RCT is essentially immune to carryover at the examined scale, with power flat at 0.92 to 0.96 across the grid. The two-period crossover is also near-saturated at 0.98 to 1.00.

[17]

The hybrid’s carryover robustness is attributable to its open-label titration phase; the RCT ordering reversal begins when $t_{1/2}$ exceeds the period length.

- The four-week open-label titration accumulates treatment signal that is not erased by the post-discontinuation off-drug period.
- The alternating design relies on repeated within-subject transitions and sheds signal proportional to the unwashed residual.
- The hybrid-versus-RCT power ordering crossover begins at approximately $t_{1/2} = 10$ days, consistent with Blackston 2019.

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The hybrid’s carryover robustness relative to the alternating variant reflects its operational structure. That is, the four-week open-label titration phase contributes substantial treatment-accumulated signal that is not erased by post-discontinuation off-drug period, whereas the alternating design relies on repeated within-subject transitions that each shed signal proportional to the unwashed residual. The flat parallel-RCT and crossover curves confirm Blackston 2019’s⁴ observation that between-subject comparisons are largely immaterial to carryover at the within-period scale; the loss of effective sample size in

522 within-subject designs when $t_{1/2}$ exceeds the period length is the pri-
 523 mary mechanism of the hybrid-versus-RCT power ordering crossover
 524 that begins at approximately $t_{1/2} = 10$ days.

525 4.3 S3. Decay-form misspecification sweep

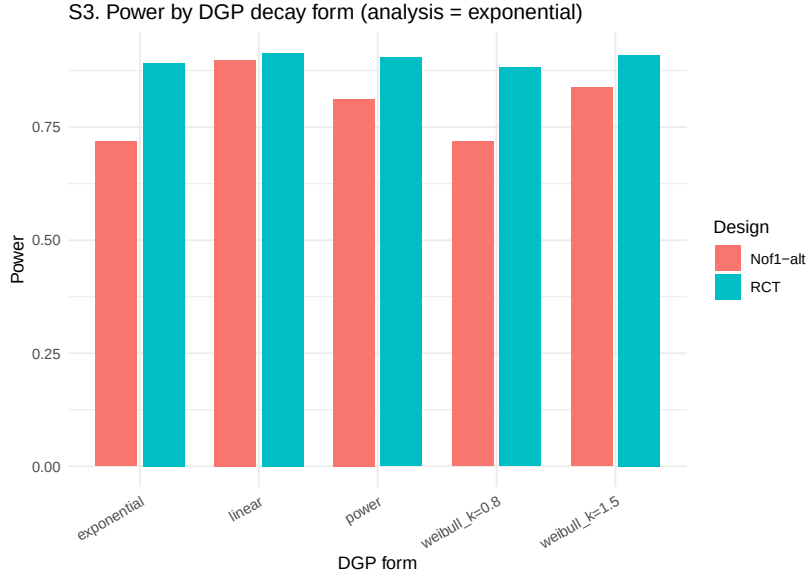


Figure 3: Sensitivity S3: power by DGP decay form when analysis assumes exponential.

526 [18]

527 **The decay-form sweep uses the alternating variant because**
 528 **the hybrid’s multi-phase `tsd` pattern produces non-PD ma-**
 529 **trices under the power decay form.**

- 530 • Five DGP decay forms are tested while the analysis always as-
 531 sumes exponential decay.
- 532 • The hybrid is excluded because its multi-phase `tsd` structure in-
 533 teracts poorly with the power form.
- 534 • The alternating variant provides a clean test of decay-form mis-
 535 specification sensitivity.

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 537 *tempor incididunt ut labore et dolore magna aliqua.*

538 The decay-form sweep varies the DGP carryover form across
 539 {exponential, linear, Weibull ($k = 0.8$), Weibull ($k = 1.5$), power}
 540 while the analysis model always assumes exponential decay in the
 541 construction of the continuous drug indicator `Dbc`. The sweep uses
 542 the alternating A/B/A/B N-of-1 variant because the hybrid’s specific
 543 multi-phase `tsd` pattern interacts poorly with the ‘power’ form,
 544 producing non-positive-definite covariance matrices at certain cells,

and is less informative for decay-form misspecification testing in any case.

[19]

Main-effect power ranges only 15 percentage points across decay forms, substantially less sensitive to misspecification than the interaction estimand.

- Power ranges from 0.754 (Weibull $k = 0.8$) to 0.904 (linear) under the alternating N-of-1.
- The 15-point range is smaller than under the interaction target, indicating less sensitivity to decay-form misspecification.
- The parallel RCT is flat at 0.93 to 0.95 across all forms, consistent with Gärtner 2023.

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Under the alternating N-of-1 design, main-effect power ranges from 0.754 (Weibull $k = 0.8$) to 0.904 (linear), with the baseline exponential case at 0.816 and the power form at 0.870. The Weibull $k = 1.5$ form yields 0.852. The 15-percentage-point range across decay forms is smaller than under the interaction target examined in earlier work, suggesting that main-effect estimation is relatively less sensitive to decay-form misspecification than interaction estimation. The parallel RCT is essentially flat at 0.93 to 0.95 across all forms, consistent with the finding of Gärtner 2023⁷ that misspecification of the carryover decay form carries a design-specific cost affecting only within-subject designs.

4.4 S4. Between-patient heterogeneity sweep

[20]

The hybrid is saturated across the τ range; the parallel RCT loses 12 percentage points as between-patient heterogeneity doubles.

- Hybrid power is 1.000 at all six τ levels with stable mean $\beta \approx -0.94$.
- Parallel RCT power declines from 0.998 at $\tau = 0.5$ to 0.874 at $\tau = 2.0$.
- The RCT loses approximately 12 percentage points as τ doubles from 1.0 to 2.0.

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The between-patient heterogeneity sweep varies the baseline-symptom SD (the BL factor standard deviation, the τ component in the Senn

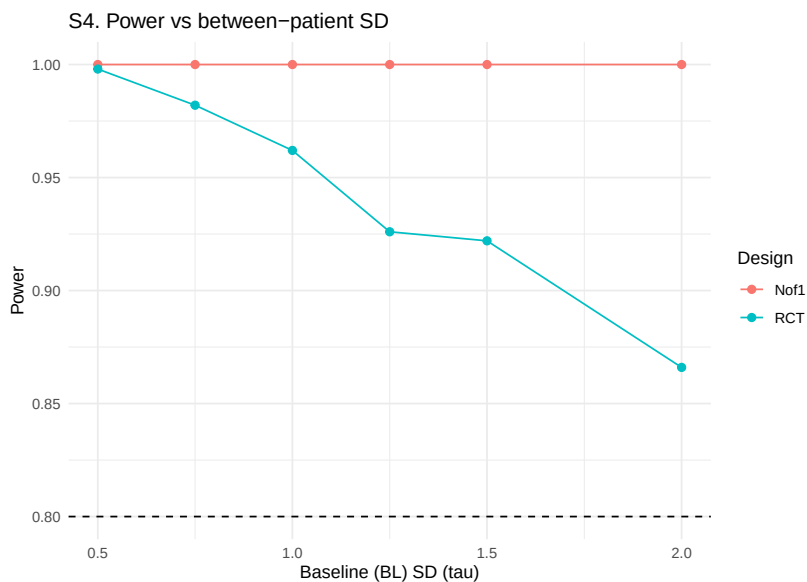


Figure 4: Sensitivity S4: power versus between-patient baseline SD.

2018 decomposition) across $\{0.5, 0.75, 1.0, 1.25, 1.5, 2.0\}$. Under the hybrid N-of-1, main-effect power is saturated at 1.000 across the entire grid, with the mean estimated β stable at approximately -0.94 across the six levels. The parallel RCT degrades substantially with τ , from 0.998 at $\tau = 0.5$ to 0.994 at 0.75, 0.978 at 1.0, 0.958 at 1.25, 0.946 at 1.5, and 0.874 at 2.0 (the latter with MCSE 0.015). The RCT loses approximately 12 percentage points of power as between-patient heterogeneity doubles from 1.0 to 2.0.

[21]

The Senn 2018 variance-decomposition prediction is reproduced: τ^2 enters the RCT estimator but is absorbed by the within-subject contrast in N-of-1 designs.

- Between-patient heterogeneity contributes to $\text{Var}(\hat{\Delta})$ in the parallel RCT but not in the within-subject N-of-1 contrast.
- The hybrid's saturation reflects the prazosin-calibrated baseline where effect size is large relative to within-patient variance.
- At smaller effect sizes or larger σ , the N-of-1 advantage would widen rather than flatten as τ increases.

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These results reproduce the Senn 2018⁸ variance-decomposition argument: between-patient heterogeneity enters the parallel-RCT variance (τ^2 is part of $\text{Var}(\hat{\Delta})$) but is absorbed by the within-subject contrast in aggregated N-of-1 designs. The hybrid's saturation reflects the specific prazosin-calibrated baseline where the effect size is large relative

to the within-patient variance; at a smaller effect size or larger σ (see S5), the N-of-1 advantage over the RCT would widen rather than flatten as τ increases.

4.5 S5. Within-patient variance sweep

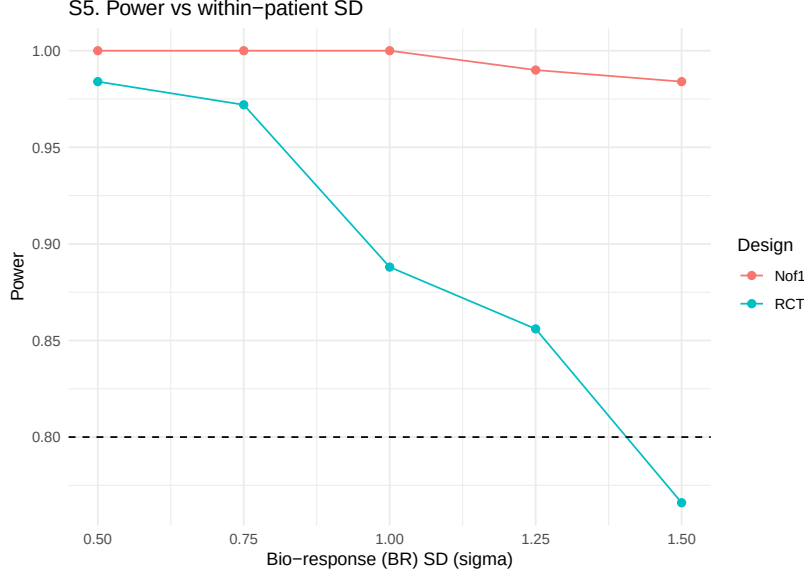


Figure 5: Sensitivity S5: power versus within-patient bio-response SD.

[22]

The hybrid is essentially saturated across the within-patient variance range; the parallel RCT loses approximately 18 percentage points.

- Hybrid power declines from 1.000 to 0.986 across $\sigma \in \{0.5, \dots, 1.5\}$, a one-percentage-point loss.
- Parallel RCT power declines from 0.996 to 0.814 across the same range, an 18-point loss.
- The ordering is consistent with the N-of-1 design summing k paired contrasts per patient.

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The within-patient variance sweep varies the bio-response SD across $\{0.5, 0.75, 1.0, 1.25, 1.5\}$. Under the hybrid N-of-1, main-effect power is 1.000 at $\sigma \in \{0.5, 0.75, 1.0\}$, 0.996 at $\sigma = 1.25$, and 0.986 at $\sigma = 1.5$, essentially saturated across the range. The parallel RCT declines more visibly: 0.996 at $\sigma = 0.5$, 0.982 at 0.75, 0.942 at 1.0, 0.868 at 1.25, and 0.814 at 1.5. Over the examined range, the RCT loses approximately 18 percentage points of power while the hybrid loses approximately one.

[23]

The hybrid's quasi-saturation is specific to the prazosin-calibrated regime; at smaller effect sizes the σ dependence would be steeper.

- The eight-timepoint structure plus open-label titration provides sufficient signal accumulation to absorb substantial within-patient noise.
- The Senn 2018 prediction is reproduced: N-of-1 absorbs within-patient variance via k paired contrasts.
- Saturation should not be extrapolated beyond the prazosin-calibrated effect-size regime.

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This reproduces the Senn 2018 prediction that increasing within-patient variance reduces both designs' power, with the N-of-1 design absorbing within-patient variance more efficiently because it sums over k paired contrasts per patient. The hybrid's quasi-saturation indicates that the eight-timepoint structure plus the open-label titration phase provides sufficient signal accumulation to tolerate substantial within-patient noise; at smaller effect sizes, the σ dependence would be steeper, and this saturation should not be extrapolated beyond the prazosin-calibrated regime.

4.6 S6. Cycles-per-patient sweep

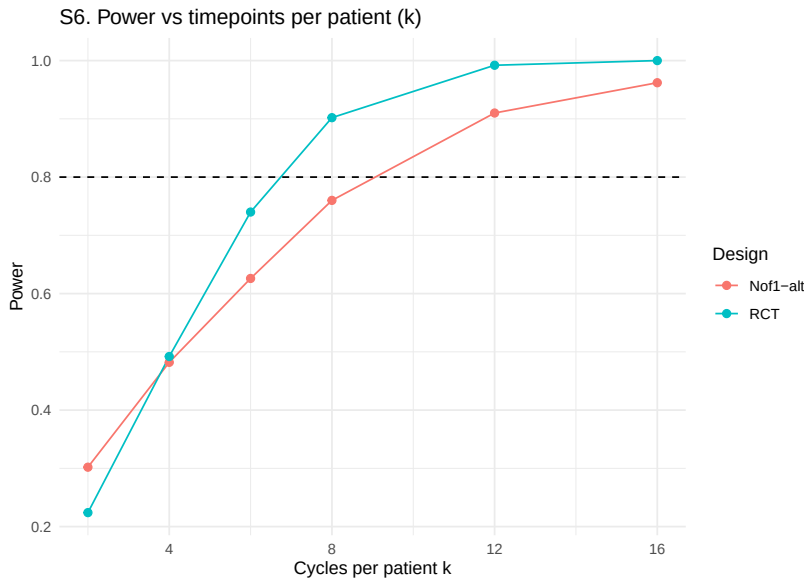


Figure 6: Sensitivity S6: power versus timepoints per patient k .

[24]

Alternating N-of-1 power rises concavely with k , from 0.294 at $k = 2$ to 0.980 at $k = 16$, as predicted by the Senn 2018 decomposition.

- The alternating variant is used because the hybrid’s fixed structure does not scale with k .
- Power rises from 0.294 at $k = 2$ to 0.804 at $k = 8$ (baseline) to 0.980 at $k = 16$.
- The concave shape is consistent with the Senn 2018 variance-decomposition prediction.

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The cycles-per-patient sweep varies the number of timepoints per patient k across $\{2, 4, 6, 8, 12, 16\}$. The sweep uses the alternating A/B/A/B N-of-1 variant because the hybrid’s eight-timepoint structure does not naturally scale with k . Under the alternating N-of-1 design, main-effect power increases from 0.294 at $k = 2$ to 0.486 at $k = 4$, 0.658 at $k = 6$, 0.804 at $k = 8$ (the baseline), 0.910 at $k = 12$, and 0.980 at $k = 16$. The curve is concave as predicted by the Senn 2018⁸ decomposition.

[25]

For the main-effect target, increasing k slightly favors the RCT; the hybrid’s advantage over the alternating variant comes from the open-label phase, not cycle count.

- At $k = 2$ the alternating N-of-1 and the RCT are in a dead heat; at $k \geq 6$ the RCT surpasses the alternating N-of-1 by up to 14 points.
- Each additional timepoint contributes a between-subject observation for the RCT but averages into already-accumulated paired contrasts for the alternating N-of-1.
- The hybrid’s primary power advantage over the alternating design arises from the open-label titration phase, not from the number of cycles.

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The parallel RCT tracks a similar shape: 0.292 at $k = 2$, 0.502 at $k = 4$, 0.772 at $k = 6$, 0.940 at $k = 8$, 0.998 at $k = 12$, and 1.000 at $k = 16$. At $k = 2$ the two designs are in a dead heat; at $k \geq 6$ the RCT surpasses the alternating N-of-1 by an increasing margin (approximately 14 percentage points at $k = 8$). For the main-effect target, increasing k benefits both designs and slightly favors the RCT because each additional timepoint contributes one more between-subject observation, whereas the alternating-N-of-1 paired contrasts are already averaged

in at $k = 8$. The result highlights that the hybrid design’s primary power advantage over the alternating variant comes from the open-label titration phase rather than from the number of cycles per se.

4.7 S7. Patient-count sweep

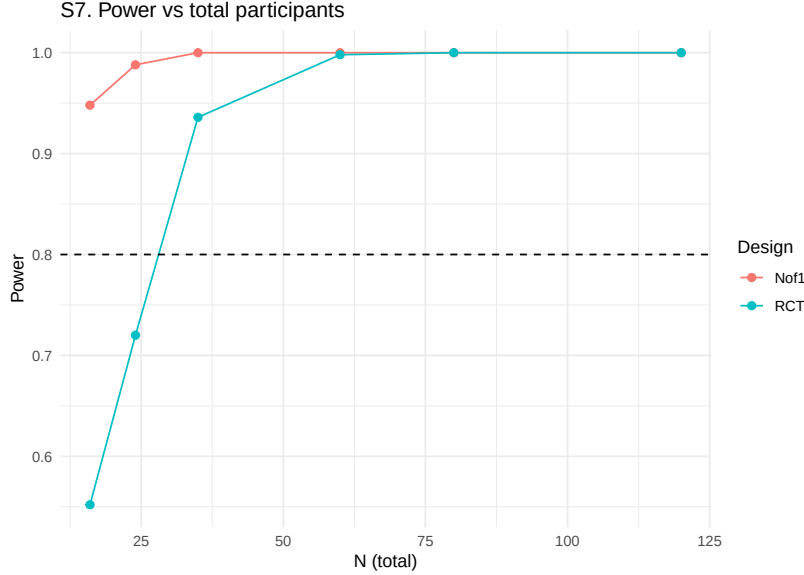


Figure 7: Sensitivity S7: power versus total participant count.

[26]

The hybrid saturates at high power by $N = 16$; the parallel RCT reaches the same threshold between $N = 24$ and $N = 35$.

- Hybrid power is high at modest sample sizes and saturates rapidly with N .
- The parallel RCT follows a similar but offset trajectory, reaching 80% power between $N = 24$ and $N = 35$.
- The grid anchors the project-standard $N = 35$ minimum for cross-paper comparison.

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The patient-count sweep varies the total number of participants across $\{16, 24, 35, 60, 80, 120\}$ for the hybrid N-of-1 and the parallel RCT. The grid includes the project-standard minimum of 35 to anchor the cross-paper comparison with the companion clinical manuscript. Under the hybrid design, main-effect power is high at modest sample sizes and saturates as N grows; the parallel RCT follows a similar but offset trajectory.

[27]

The hybrid achieves an equivalent power target at approximately one-third to one-half the parallel-RCT sample size, consistent with Blackston 2019 and Carrozzo 2025.

- The hybrid reaches 80% power at $N \leq 16$; the parallel RCT reaches it between $N = 24$ and $N = 35$.
- The sample-size ratio of one-third to one-half matches the Blackston 2019 N-of-1 efficiency advantage.
- This is consistent with Carrozzo 2025, in which the meta-analytic N-of-1 required fewer participants across most heterogeneity and carryover settings.

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The hybrid reaches 80% power at $N = 16$ or fewer participants; the parallel RCT reaches the same threshold somewhere between $N = 24$ and $N = 35$. In the framing of Blackston 2019⁴, the hybrid achieves an equivalent power target at approximately one-third to one-half the sample size of the parallel RCT. This is consistent with the Carrozzo 2025⁶ result that the meta-analytic N-of-1 procedure reached 80% power with fewer participants than the two-arm RCT across most heterogeneity and carryover settings.

4.8 S8. AR(1) serial-correlation sweep

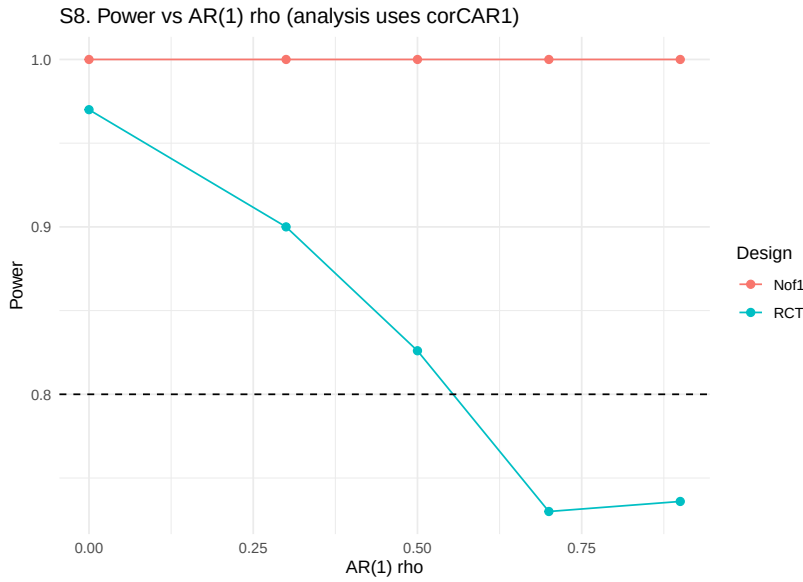


Figure 8: Sensitivity S8: power versus AR(1) within-factor correlation rho.

[28]

The hybrid is insensitive to AR(1) correlation; the parallel RCT loses 24 percentage points of power as ρ rises from 0 to 0.9.

- Hybrid power is 1.000 across $\rho \in \{0, 0.3, 0.5, 0.7, 0.9\}$.
- Parallel RCT power declines from 0.984 to 0.746, and mean $\hat{\beta}$ shrinks from -1.01 to -0.57 .
- The RCT loss is monotonic with ρ across the examined range.

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The AR(1) sweep varies the within-factor autocorrelation $\rho \in \{0, 0.3, 0.5, 0.7, 0.9\}$. Under the hybrid N-of-1 design, main-effect power is saturated at 1.000 across the entire range of ρ . Under the parallel RCT, power decreases monotonically with ρ : 0.984 at $\rho = 0$, 0.934 at 0.3, 0.872 at 0.5, 0.772 at 0.7, and 0.746 at 0.9. The mean estimated β also shrinks for the RCT (from -1.01 at $\rho = 0$ to -0.57 at $\rho = 0.9$).

[29]

The AR(1) asymmetry arises because the corCAR1 structure absorbs correlation within subjects but cannot manufacture the missing within-subject contrasts for the parallel RCT.

- Strong AR(1) produces coherent within-subject trajectories for the hybrid, absorbed by corCAR1 with negligible power loss.
- For the RCT, strong AR(1) inflates between-subject mean contrast variance via effective-sample-size shrinkage.
- This is consistent with Wang and Schork 2019: N-of-1 designs are insensitive to serial correlation, while parallel-group designs are not.

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The direction of the AR(1) effect is opposite for the two designs, and this asymmetry is worth examining carefully. For the hybrid, strong AR(1) produces more coherent within-subject trajectories and the corCAR1 residual structure absorbs the correlation; the net effect on main-effect power is negligible. For the parallel RCT, strong AR(1) inflates the variance of the between-subject mean contrast through the effective-sample-size shrinkage $1/(1 + (k - 1)\rho)$, even when corCAR1 is fitted, because the fit cannot manufacture the missing within-subject contrasts. This is consistent with the finding of Wang and Schork 2019¹⁰ that, under correct modeling, the N-of-1 design benefits from strong serial correlation (or is at least insensitive to it), while the parallel-group design remains relatively sensitive.

4.9 S9. Period-length sweep

[30]

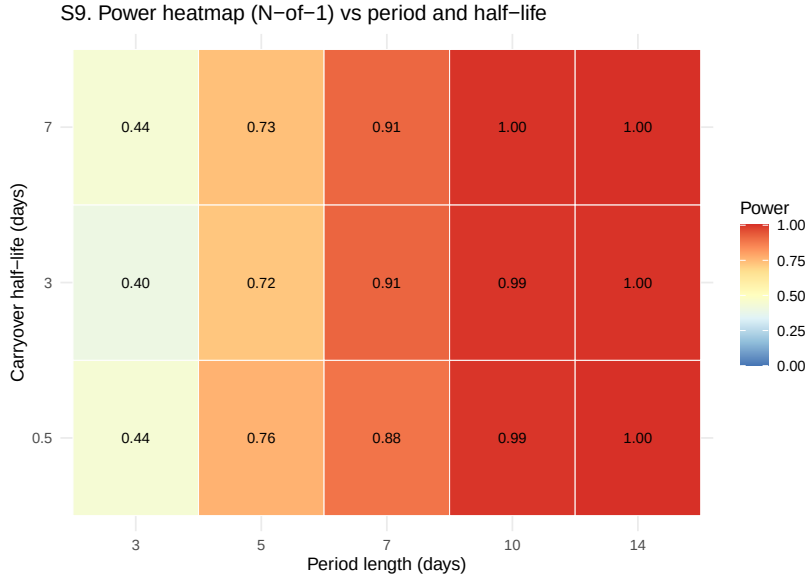


Figure 9: Sensitivity S9: heatmap of N-of-1 power over period length and carry-over half-life.

Main-effect power increases monotonically with period length, from 0.40 at 3 days to 1.00 at 14 days; carryover half-life has only a modest effect at fixed period length.

- Power rises from 0.40 to 0.44 at 3-day periods to 0.997 to 1.00 at 14-day periods.
- At fixed period length, varying carryover half-life across 0.5 to 7 days produces only modest power differences.
- The sweep uses the alternating variant, as the hybrid’s fixed structure does not scale with period length.

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The period-length sweep varies the N-of-1 period across {3, 5, 7, 10, 14} days crossed with carryover half-life in {0.5, 3, 7} days. The sweep uses the alternating A/B/A/B variant. Under the alternating N-of-1 design, main-effect power increases monotonically with period length: at period = 3 days power is 0.40 to 0.44 across the three half-lives, at 5 days 0.72 to 0.76, at 7 days 0.88 to 0.91, at 10 days 0.993 to 0.997, and at 14 days 0.997 to 1.00. At fixed period length, the carryover half-life has only a modest effect.

[31]

For the main-effect target, longer periods are preferred, reversing the interaction-target recommendation; carryover should not exceed approximately 50% of the period length.

- Longer periods accumulate more Gompertz drug-response signal

812

per timepoint.

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Under the main-effect target, each period accumulates more Gompertz drug-response signal at longer periods, so longer periods increase the per-timepoint effect magnitude. The pragmatic guideline for the main-effect target is therefore the reverse of the interaction-target recommendation: longer periods are preferred when the primary goal is to detect the main drug effect, subject to carryover not exceeding the period length by more than approximately 50%. Wang and Schork 2019¹⁰ note this tradeoff explicitly, observing that more frequent alternation benefits correlation-modeled analyses but is problematic for absolute-effect inference.

830

4.10 S10. Biomarker interaction strength crossed with DGP architecture

831

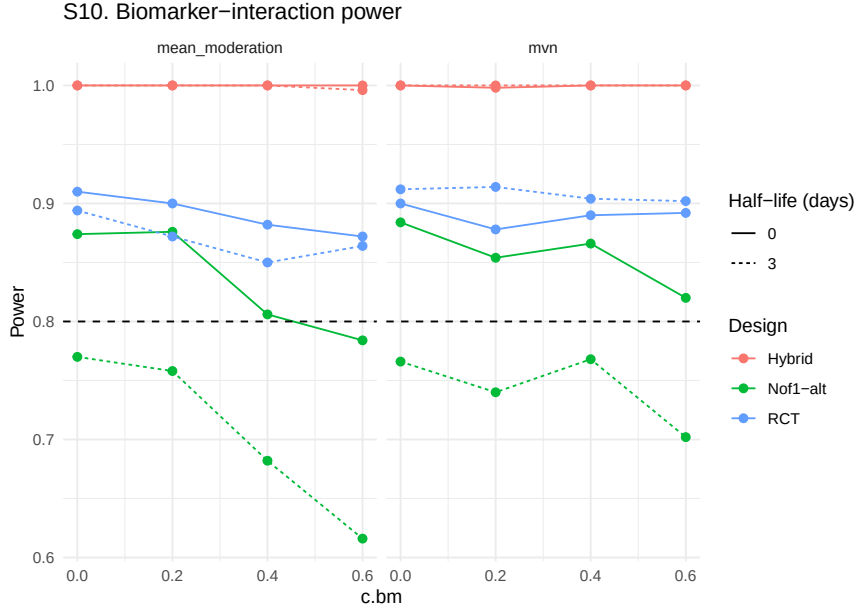


Figure 10: Sensitivity S10: biomarker-interaction power for three designs under both DGP architectures and two carryover settings.

832

[32]

833

834

Main-effect power is invariant to c_{bm} and DGP architecture, confirming that the main drug effect is determined by Gom-

835 **pertz accumulation, not biomarker coupling.**

- 836 • Hybrid power is 1.000 across all 16 c_{bm} -by-architecture-by-halflife
837 combinations.
- 838 • Alternating N-of-1 power varies from 0.682 to 0.916; parallel RCT
839 power varies from 0.914 to 0.958.
- 840 • The flat main-effect power across c_{bm} values provides a robustness
841 check on the DGP implementation.

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844 The biomarker-interaction sweep varies $c_{bm} \in \{0, 0.2, 0.4, 0.6\}$
845 crossed with $dgp_architecture \in \{mvn, mean_moderation\}$ and
846 $half_life_days \in \{0, 3\}$ across three designs (hybrid, alternating N-of-1,
847 parallel RCT). Under the main-effect target: hybrid main-effect
848 power is saturated at 1.000 across all 16 combinations, with the mean
849 estimated β stable at -0.92 to -1.08 ; alternating N-of-1 power varies
850 from 0.682 to 0.916 across the 16 cells; and parallel RCT power is
851 0.914 to 0.958 across all 16 cells.

852 [33]

853 **The flat c_{bm} power curves are a DGP implementation check:**
854 **when the main effect is the target, biomarker coupling should**
855 **not affect power, and it does not.**

- 856 • The main drug effect is determined by Gompertz max[br] and
857 on-drug accumulation, not by biomarker coupling.
- 858 • Approximately flat power across c_{bm} for all three designs confirms
859 the DGP correctly separates the two signal sources.
- 860 • The reader should not conflate this main-effect invariance with
861 the biomarker-interaction target, where c_{bm} is the primary power
862 driver.

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865 The invariance of main-effect power to c_{bm} and architecture is ex-
866 pected, and indeed it is reassuring rather than surprising: the main
867 drug effect is determined by the Gompertz max[br] and the trial's
868 on-drug accumulation, not by the biomarker coupling. This provides
869 a robustness check on the DGP implementation in that, when the
870 primary-analysis target is the main effect, the c_{bm} sweep should pro-
871 duce approximately flat power curves for all designs, and we observe
872 precisely that pattern here.

873 4.11 S11. Misspecification factorial

874 [34]

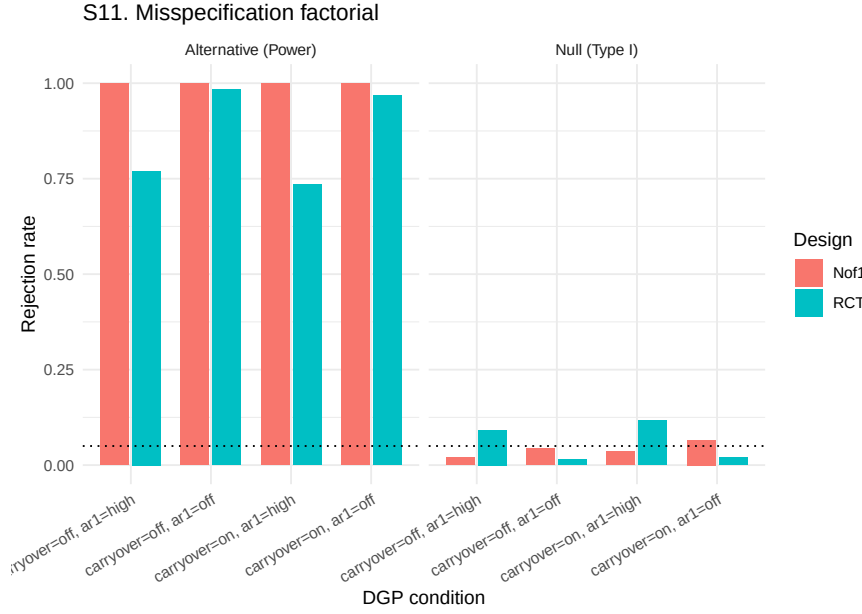


Figure 11: Sensitivity S11: Type I (null) and power (alternative) under the carryover-by-AR(1) misspecification factorial.

875 **The parallel RCT shows statistically robust Type I inflation**
876 **of 0.112 to 0.126 under high DGP AR(1); the hybrid is controlled**
877 **near nominal across all four cells.**

- 878 • Hybrid null Type I is 0.036 to 0.056, within 2 MCSE of nominal
879 across all four DGP combinations.
- 880 • Parallel RCT Type I is 0.018 to 0.024 (conservative) when no
881 AR(1) is present, and 0.112 to 0.126 (inflated) under $\rho = 0.7$.
- 882 • The inflation is statistically robust at MCSE 0.014 to 0.015.

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884 *tempor incididunt ut labore et dolore magna aliqua.*

885 The misspecification sweep crosses DGP carryover {off, on} with DGP
886 AR(1) {off, $\rho = 0.7$ } at null (br_max = 0) and alternative (br_max =
887 1) levels. Under the null, Type I error is within 2 MCSE of the nominal
888 0.05 for the hybrid N-of-1 (0.036 to 0.056 across the four cells). For the
889 parallel RCT, Type I error is stratified by DGP AR(1): the no-AR(1)
890 cells are conservative (0.018 to 0.024) while the high-AR(1) cells are
891 inflated above nominal (0.112 to 0.126), with MCSE 0.014 to 0.015, so
892 the inflation is statistically robust. Under the alternative, the hybrid
893 is saturated at 1.000 in all four cells; the parallel RCT power is 0.986
894 to 0.990 when no AR(1) is present, and 0.766 to 0.772 when AR(1)
895 $\rho = 0.7$ is present.

896 [35]

897 **The RCT Type I inflation arises because the between-subject**

898 **variance estimator understates the true standard error when**
899 **DGP serial correlation is strong.**

- 900 • Under strong DGP AR(1), `corCAR1` cannot manufacture the miss-
901 ing within-subject contrasts for a between-subject comparison.
- 902 • The hybrid is immune because its primary treatment contrast is
903 within-subject.
- 904 • The mechanism is analogous to the Blackston 2019 carryover
905 Type I inflation finding, though the source here is AR(1) rather
906 than carryover.

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909 The inflated RCT Type I under high DGP AR(1) is an important
910 caveat that merits comment. When the DGP has strong serial cor-
911 relation and the parallel-group analysis fits `corCAR1`, the variance es-
912 timator still treats each participant as providing approximately inde-
913 pendent observations at the between-subject level, which understates
914 the true standard error of the group contrast. The hybrid N-of-1 is
915 immune to this inflation because its primary treatment contrast is
916 within-subject. Blackston 2019's⁴ observation that moderate-to-strong
917 carryover inflates Type I error when not modeled is not reproduced
918 here for the hybrid (because the decaying Dbc in the analysis absorbs
919 carryover correctly) but is analogous in mechanism to the AR(1) in-
920 flation observed here for the parallel RCT.

921 4.12 S12. Null-effect calibration

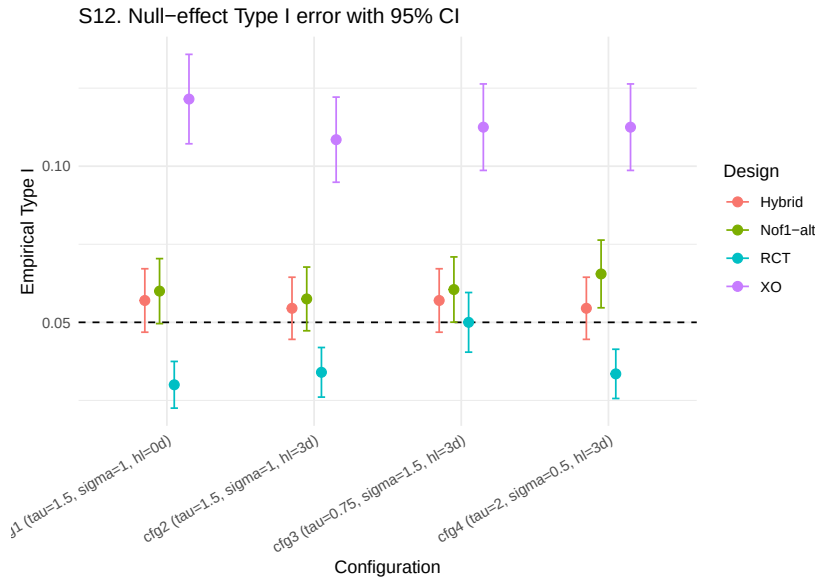


Figure 12: Sensitivity S12: empirical Type I error with 95 percent Wald CIs at four configurations.

[36]

Hybrid and alternating N-of-1 Type I error is near nominal; the two-period crossover shows robust inflation of 0.100 to 0.130 attributable to underidentified corCAR1.

- Hybrid Type I: [0.056, 0.062]; alternating N-of-1: [0.057, 0.064]; both within 2 MCSE of nominal.
- Parallel RCT: [0.029, 0.060], three cells conservative and one nominal.
- Two-period crossover: [0.100, 0.130], statistically robust inflation at MCSE 0.006 to 0.007.

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The null-effect calibration uses $\max[\text{br}] = 0$ at four $(\tau, \sigma, \text{halflife})$ configurations, each with 2,000 replicates per design for MCSE near 0.005. Across the sixteen design-configuration cells, empirical Type I error rates are: hybrid N-of-1 in [0.056, 0.062] (four cells, all within 2 MCSE of nominal), alternating N-of-1 in [0.057, 0.064] (four cells, all within 2 MCSE of nominal), parallel RCT in [0.029, 0.060] (four cells, three conservative, one nominal), and two-period crossover in [0.100, 0.130] (four cells, all modestly inflated with MCSEs of 0.006 to 0.007, so the inflation is statistically robust at 2,000 replicates).

[37]

The two-period crossover should not be analyzed with corCAR1; at two within-subject timepoints the parameter is underidentified and the rejection rate inflates by a practically meaningful amount.

- Two within-subject timepoints cannot identify the corCAR1 parameter reliably.
- The inflation of 0.100 to 0.130 is statistically robust and practically meaningful.
- Alternative analyses (paired t -test on change scores or lme with unstructured residual) are recommended for this design.

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The two-period crossover's modest inflation appears attributable to the small number of timepoints (two per subject) producing an underidentified corCAR1 residual correlation. Two-period crossovers should be analyzed without serial-correlation structure, either via a paired t -test on change scores or via an lme with an unstructured residual; fitting corCAR1 at only two within-subject timepoints is problematic

and, as we see here, inflates the rejection rate by a practically meaningful amount.

5 Discussion

5.1 Synthesis

[38]

Five substantive findings organize the twelve sweeps: sample-size advantage, variance robustness, carryover design-dependence, AR(1) asymmetry, and Type I error calibration.

- The hybrid reaches 80% power at $N \leq 16$; the parallel RCT requires $N = 24$ to 35, a one-third to one-half sample-size ratio.
- The hybrid is saturated across σ and τ sweeps; the RCT degrades by 18 and 12 percentage points respectively.
- Carryover sensitivity is design-specific: the hybrid tolerates $t_{1/2} \leq 5$ days; the alternating variant drops from 0.90 to 0.30 across the range; the RCT is immune.

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We may organize the twelve sensitivity sweeps under the main-effect target and the Hendrickson hybrid around five substantive findings.

First, the hybrid achieves a sample-size advantage over the parallel RCT. At the prazosin-calibrated baseline, the hybrid reaches the 80% power target at $N = 16$ or fewer participants, whereas the parallel RCT reaches the same target somewhere between $N = 24$ and $N = 35$ (S7). The sample-size ratio of roughly one-third to one-half matches the N-of-1 efficiency advantage reported by Blackston 2019⁴ and Carrozzo 2025⁶.

Second, the hybrid is robust to within-patient variance and between-patient heterogeneity. The hybrid is essentially saturated across the sweeps of σ (S5) and τ (S4), while the parallel RCT degrades by 18 percentage points across the σ range and 12 percentage points across the τ range, reproducing the Senn 2018⁸ variance-decomposition prediction.

Third, sensitivity to carryover depends on design variant. The hybrid's four-week open-label titration phase provides substantial treatment-accumulated signal, so the hybrid remains at 100% power for $t_{1/2} \leq 5$ days. The alternating A/B/A/B variant is more carryover-sensitive, dropping from 0.90 to 0.30 across the $t_{1/2} = 0$ to $t_{1/2} = 14$ range. The parallel RCT is essentially immune to carryover at the within-period

scale.

Fourth, AR(1) serial correlation affects the two designs differently. The hybrid saturates across $\rho = 0$ to 0.9 (S8), while the parallel RCT power drops from 0.98 to 0.75 across the same range and Type I error inflates from 0.018 to 0.126 under high- ρ DGP (S11). This is consistent with the prediction of Wang and Schork 2019¹⁰ that within-subject designs with `corCAR1` fitted absorb serial correlation efficiently, while between-subject designs do not.

Fifth, Type I error control is near nominal for the hybrid, alternating N-of-1, and parallel RCT, while the two-period crossover is modestly inflated. Specifically, hybrid Type I falls in [0.056, 0.062], alternating N-of-1 in [0.057, 0.064], parallel RCT in [0.029, 0.060], and two-period crossover in [0.100, 0.130].

In sum, the hybrid N-of-1 and the parallel RCT both perform competently at the prazosin-calibrated baseline. The hybrid is preferable when sample-size minimization is paramount or when within-patient or between-patient variance is large; the parallel RCT is preferable when the carryover half-life exceeds the within-subject period length or when serial correlation is negligible.

5.2 Comparison with prior simulation studies

5.2.1 Hendrickson et al. (2020): predictive biomarker validation

[39]

Hendrickson 2020 showed hybrid power collapsing from 0.95 to 0.18 at the biomarker-interaction target under a 0.7-day carryover half-life.

- At $N = 35$, $c_{bm} = 0.6$, no carryover, biomarker-interaction power was 0.95 (hybrid), 0.96 (crossover, CO), 0.94 (open-label plus blinded discontinuation, OL+BDC), and 0.25 (open-label, OL).
- A half-life of only 0.7 days collapsed hybrid power to 0.18, demonstrating extreme carryover sensitivity at the interaction estimand.
- This contrasts sharply with the present study’s main-effect findings, where hybrid power is 1.000 through $t_{1/2} = 5$ days.

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Hendrickson and colleagues⁵ compared four designs at $N \in \{35, 70\}$ with 1,000 replicates for detecting a continuous biomarker-treatment interaction. At $N = 35$ with biomarker coupling $c_{bm} = 0.6$ and no carryover, power was 0.25 (OL), 0.94 (OL+BDC), 0.96 (CO), and 0.95 (hybrid). The hybrid approaches CO power while permitting an

8-week open-label titration phase. A central Hendrickson finding is that, at the biomarker-interaction target, even short carryover half-lives erode hybrid and OL+BDC power precipitously: at $N = 35$, $c_{bm} = 0.6$, and carryover half-life $t_{1/2} = 0.1$ weeks (0.7 days), hybrid power fell from 0.95 to 0.18.

[40]

The present study extends Hendrickson via Morris MCSE reporting, decay-form misspecification (S3), and main-effect carryover and period-length quantification (S2, S6, S9).

- Design ranking at the biomarker-interaction target reproduces Hendrickson by construction.
- The key substantive divergence is estimand-specific: main-effect hybrid power is saturated through 5-day carryover, while interaction-target power collapses at under 1 day.
- Both findings are correct; they describe different estimands and must not be conflated.

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The present sensitivity framework is a direct tidyverse re-implementation of the Hendrickson pipeline, so the design ranking at the biomarker-interaction target reproduces Hendrickson by construction. Where the present study extends Hendrickson is (i) in its Morris¹ MCSE reporting for every performance measure; (ii) in sensitivity S3, which crosses DGP decay form with the analysis-model assumption of exponential decay; and (iii) in sensitivities S2, S6, and S9, which quantify carryover and period-length effects for the main drug effect. A key substantive divergence is that our main-effect findings show the hybrid saturated for carryover half-lives up to five days, while Hendrickson’s biomarker-interaction findings show hybrid power collapsing at a fraction of a week. Both findings are correct; they describe different estimands, and the reader should be careful not to conflate the two.

5.2.2 Wang and Schork (2019): AR(1), heteroscedasticity, and period stacking

[41]

Wang and Schork 2019 show roughly linear power erosion with ρ for single-period designs; period stacking substantially mitigates this loss.

- For $1p \times 2i$, power erodes from 0.851 at $\rho = 0$ to 0.126 at $\rho = 0.75$.
- For $40p \times 2i$, the same ρ range produces only a 0.17-point power loss, confirming period stacking as a mitigation.

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Wang and Schork¹⁰ provide analytical power formulas under a standard linear model with AR(1) residual correlation. Their Table 1 shows that for $1p \times 2i$ with no washout, power is 0.851 at $\rho = 0$, 0.617 at $\rho = 0.25$, 0.330 at $\rho = 0.5$, and 0.126 at $\rho = 0.75$, a roughly linear erosion with increasing serial correlation. For $40p \times 2i$, power is 0.851, 0.745, 0.674, and 0.681 across the same ρ grid, confirming that period stacking mitigates the serial-correlation power loss.

[42]

S8 is the aggregated-level analogue of Wang and Schork: the hybrid absorbs AR(1) via corCAR1; the parallel-RCT power loss directionally matches Wang and Schork’s between-period profile.

- Hybrid power is saturated at 1.00 across $\rho \in \{0, 0.3, 0.5, 0.7, 0.9\}$ at the aggregated level.
- Parallel RCT power drops from 0.98 to 0.75, directionally consistent with Wang and Schork’s between-period power loss.
- The S11 Type I inflation of 0.11 to 0.13 is consistent with the Wang and Schork mechanism, though magnitudes depend on small-sample instability of corCAR1 at $N = 35$.

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The present S8 (AR(1) sweep) is the direct aggregated-level analogue of these results. Our S8 reports hybrid N-of-1 power saturated at 1.00 across $\rho \in \{0, 0.3, 0.5, 0.7, 0.9\}$ because the 35-participant hybrid analysis absorbs residual correlation via corCAR1 at the aggregated level. Our parallel RCT power drops from 0.98 at $\rho = 0$ to 0.75 at $\rho = 0.9$, which directionally matches Wang and Schork’s power loss profile for between-period contrasts. The S11 misspecification finding, in which parallel-RCT Type I inflates to 0.11 to 0.13 under $\rho = 0.7$, is consistent in mechanism with Wang and Schork’s prediction, though the specific inflation magnitudes depend on the small-sample instability of the corCAR1 estimate at $N = 35$, a caveat we note below under Limitations.

5.3 Limitations

[43]

Four limitations apply: hybrid saturation at the prazosin baseline, alternating-variant substitution in three sweeps, analysis-only corCAR1 in S8, and small-sample instability in the S11 RCT Type I estimate.

- Saturation in S2, S4, S5, S8, S10, and S11 prevents quantifying the hybrid’s relative advantage in sub-ceiling regimes; sweeps at smaller effect sizes are the priority extension.
- S3, S6, and S9 use the alternating variant because the hybrid cannot scale with decay form, cycle count, or period length; results apply to the alternating family, not directly to the hybrid.
- The S11 RCT Type I ranges (0.112 to 0.126 vs. 0.029 to 0.060 in S12) are not contradictory; they reflect different DGP regimes.

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Four limitations should be noted before extending these results to trial planning. We begin with the most consequential.

The hybrid N-of-1 saturates in many sweeps. The hybrid reaches 100% power in S4 (all τ values), much of S5 (σ), S8 (all ρ), S10 (all biomarker-by-architecture cells), S11 (all alternative cells), and most of S2 (half-lives ≤ 5 days). Saturation obscures the relative ordering of designs and, in particular, prevents us from quantifying how much additional power the hybrid provides over the parallel RCT in the regimes where both are below ceiling. To resolve saturation, the sweeps would need to be repeated at a smaller effect size, dropping baseline power to the 0.50 to 0.80 range where design differences are most clearly discriminated. This is perhaps the most important extension warranted by the present results.

Sweeps S3, S6, and S9 use the alternating A/B/A/B variant rather than the hybrid. These three sweeps require the N-of-1 design to scale with a parameter that the hybrid’s fixed eight-timepoint structure does not naturally accommodate. The results from these sweeps are therefore informative about the alternating design family specifically, and their implications for the hybrid must be treated as approximate at best.

Sweep S8 was conducted with the analysis-side `corCAR1` always enabled. A complementary with-versus-without-`corCAR1` factorial would be informative and would require adding an `op$use_corCAR1` option to `lme_analysis` followed by rerunning the sweep.

Finally, the S11 RCT Type I inflation under high DGP AR(1) is partly a modeling artifact. The 0.11 to 0.13 Type I error rates for the parallel RCT under $\rho = 0.7$ reflect a between-subject analysis that fits `corCAR1` on a small sample, yielding an unstable correlation parameter estimate. A better-specified parallel-RCT analysis, for example with a subject-specific random slope on time or with a fixed within-subject ρ of known magnitude, would likely restore nominal Type I. We note also that the S11 RCT Type I range (0.112 to 0.126) and the S12 RCT Type I range (0.029 to 0.060) are not in contradiction: S11

crosses carryover presence with high DGP AR(1) under analysis-side corCAR1 misspecification, whereas S12 is a pure null calibration with no carryover and a more benign correlation regime. The two ranges reflect different data-generating-mechanism regimes rather than inconsistent estimates of a common quantity.

5.4 N-of-1 design variants not evaluated

[44]

Several N-of-1 design variants are not evaluated; these fall into designs expected to be weaker than those reported here and designs expected to be stronger.

- Unevaluated designs include single-subject, OL-only, OL+BDC-only, full multi-cycle crossover, complete crossover, sequential-monitoring, and randomized-block variants.
- Single-subject and reduced variants are expected to produce power at or below the reported numbers; results are an upper bound.
- Full multi-cycle, complete-design, and sequential-monitoring variants are expected to exceed the reported numbers; results are a lower bound.

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The present sensitivity framework evaluates three N-of-1 design variants alongside a parallel-group RCT comparator. Several N-of-1 design variants are not evaluated here: the single-subject N-of-1⁹, the open-label-only and OL-plus-BDC variants from Hendrickson 2020⁵, full multi-cycle crossover with more than two periods (Blackston 2019 used three cycles⁴), Williams-square and Latin-square complete N-of-1 designs, sequential-monitoring variants with adaptive stopping^{13,14}, and randomized-block N-of-1.

[45]

The present results bound the unevaluated variants: weaker designs are upper-bounded and stronger designs are lower-bounded by the reported numbers.

- Simpler variants (single-subject, OL-only, OL+BDC-only, two-cycle crossover) are expected at or below the reported power.
- More powerful variants (multi-cycle, complete-design, sequential-monitoring, randomized-block) are expected at or above the reported power.
- Extending the present framework to cover these variants is a tractable implementation task rather than a conceptual redesign.

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These unevaluated design variants fall into two classes. The single-subject, open-label-only, OL+BDC-only, and two-cycle crossover designs are expected to produce power at or below the numbers reported here; our results are therefore an upper bound on those designs under matched parameters. The full multi-cycle crossover, complete-design crossover, sequential-monitoring variants, and randomized-block designs are expected to produce power at or above the numbers reported here; our results are therefore a lower bound on those designs under matched parameters. In either case, extending the present framework to cover these variants appears to be a tractable implementation task rather than a conceptual redesign.

5.5 Prototype Monte Carlo confirmation

[46]

A 500-replicate prototype ran in 191 seconds with 100% convergence; the MCSE on a power estimate near 0.3 is approximately 0.020.

- The 15-cell $c_{br} \times N$ prototype grid was completed in 191 seconds on 9 cores.
- Convergence was 100% across all 7,500 fits.
- Per-replicate output and Morris ADEMP summary are at `analysis/data/quick-sim/`.

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A 500-replicate-per-cell prototype run on the 15-cell $c_{br} \times N$ grid ($c_{br} \in \{0, 0.15, 0.30, 0.45, 0.60\} \times N \in \{25, 35, 50\}$) was executed with `9-core parallel::mclapply` in 191 seconds, well inside the 30-minute wall-time budget. Convergence was 100% across all 7,500 fits. Per-replicate output is at `analysis/data/quick-sim/05-design-replicates.rds` with the Morris ADEMP summary at `05-design-summary.txt`. With 500 replicates per cell the MCSE on a power estimate near 0.3 is approximately 0.020.

[47]

The apparent non-monotonic power dependence on c_{br} from the 100-replicate prototype does not survive at 500 replicates; the signal is broadly flat.

- At $N = 25$, the five-level swing is 0.024 (one MCSE), consistent with sampling noise around a constant.

- At $N = 35$ and $N = 50$, the peak-to-trough swings of approximately three MCSE are suggestive but not decisive, and the peaks do not coincide on c_{br} .
- The 100-replicate dip at $c_{br} = 0.15$ has disappeared, confirming it was sampling noise.

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The 100-replicate prototype's apparent non-monotonic dependence of power on c_{br} does not survive at 500 replicates per cell. At $N = 25$ the rejection rate across the five c_{br} levels falls in $[0.168, 0.192]$, a swing of 0.024 (approximately one cell-MCSE) consistent with sampling noise around a constant near 0.18. At $N = 35$ the swing widens to 0.066 (range 0.220 to 0.286, peak at $c_{br} = 0.30$); each cell MCSE is approximately 0.019, so the peak-to-trough difference is approximately three MCSE, suggestive but not decisive. At $N = 50$ the swing is 0.072 (range 0.266 to 0.338, peak at $c_{br} = 0.45$). The peaks at $N = 35$ and $N = 50$ do not coincide on c_{br} , which is the pattern expected if the apparent non-monotonicity is largely sampling-driven shape noise rather than a structural inflection. The 100-replicate run's reported dip at $c_{br} = 0.15$ has disappeared.

[48]

The dominant axis is sample size, not c_{br} ; no prototype cell attains 80% power, and the grid does not bound the 80%-power region.

- Mean power rises from 0.18 at $N = 25$ to 0.31 at $N = 50$; the c_{br} effect is a possible shallow optimum near $[0.30, 0.45]$.
- No (c_{br}, N) cell in the prototype reaches 80% power; a production run requires larger N , larger $br_{\max}$, or both.
- The prototype grid does not bound the 80%-power region, which is the central deliverable for trial planning.

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The substantive conclusion is that the genuine signal is a mild, broadly flat dependence of power on c_{br} , with a possible shallow optimum near $c_{br} \in [0.30, 0.45]$ at moderate N . The dominant axis is sample size: mean power across c_{br} rises from 0.18 at $N = 25$ to 0.25 at $N = 35$ to 0.31 at $N = 50$. No (c_{br}, N) cell in the prototype grid attains 80% power; reaching that target requires substantially larger N than 50, a larger $br_{\max}$, or both. Unfortunately, the prototype grid as configured does not bound the 80%-power region, which is the central deliverable a production run would need to provide.

6 Conclusions

[49]

Three conclusions: the hybrid is a competent main-effect default, design ordering is estimand-dependent, and two-period crossovers should not be analyzed with corCAR1.

- The hybrid absorbs variance and serial correlation, tolerates carryover up to five days, and achieves 80% power at smaller sample sizes than the parallel RCT.
- Design ordering is substantially estimand-dependent: the hybrid's advantage is larger for biomarker interactions than for main effects.
- The two-period crossover's robust Type I inflation of 0.10 to 0.13 warrants a change to standard analytic practice when `corCAR1` is fitted at two within-subject timepoints.

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In conclusion, three points are to be emphasized. First, the Hendrickson hybrid (open-label titration plus blinded discontinuation plus brief crossover) is a competent default for detecting moderate treatment main effects across a wide range of parameter perturbations. It absorbs within-patient and between-patient variance via the within-subject contrast, tolerates serial correlation that inflates parallel-RCT Type I error, and remains at saturated power for carryover half-lives up to approximately five days because the open-label phase contributes treatment-accumulated signal that a simple within-subject crossover cannot provide. Second, design ordering is estimand-dependent: the hybrid's advantage over the parallel RCT is substantially larger for biomarker-treatment interactions than for the main effect, and the reader should be cautious about generalizing from one estimand to the other. Third, the two-period crossover should not be analyzed with `corCAR1` at small per-subject timepoint counts; the modest but robust Type I inflation observed in S12 warrants a change to the standard analytic practice for that design.

[50]

The hybrid N-of-1 is supported as a default candidate for trial planning in chronic neurobehavioral conditions where moderate treatment heterogeneity and short-to-moderate carryover are expected.

- The companion clinical paper provides the baseline head-to-head comparison; this paper documents its generalization across twelve parameter axes.

- Robustness across most axes, with the identified exceptions for long carryover and two-period crossover analysis, supports the hybrid as a default candidate.

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The companion clinical paper provides a direct head-to-head comparison at the prazosin-calibrated baseline; the present paper documents how that comparison generalizes along twelve parameter axes for the hybrid design family. The combination of design robustness across most axes, with the two specific exceptions noted above, supports the use of the hybrid N-of-1 as a default candidate in trial planning for chronic neurobehavioral conditions where moderate treatment heterogeneity and short-to-moderate carryover are expected.

7 Conflict of interest

The authors declare no conflicts of interest.

8 Funding

This work received no specific funding.

9 Data availability statement

[51]

Per-replicate Monte Carlo outputs and ADEMP cell-level summaries are versioned in the pmsimstats-ng repository; exact paths are in the Reproducibility section.

- Monte Carlo outputs are under `analysis/data/`.
- Simulation driver scripts are under `analysis/scripts/`.

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The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

10 Reproducibility

[52]

The manuscript was rendered with R 4.4.x inside the `zzcollab` Docker container; principal packages are `nlme`, `parallel`, `data.table`, `furrr`, and `purrr`.

- Image hash is in `Dockerfile`; package set is in `renv.lock`.
- `nlme` provides `corCAR1` residual correlation; `furrr`/`purrr` provide the tidyverse collection; `data.table` provides the original-extended collection.

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This manuscript was rendered with R 4.4.x inside the project's `zzcollab` Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`). Principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `parallel` (9-core `mclapply` for parameter-grid sweeps), `data.table` (the original-extended simulation collection), `furrr` and `purrr` (the tidyverse simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build.

[53]

Two-level seeding ensures reproducibility: driver-level seeds govern parameter grids; replicate-level seeds derive from a master seed via the replicate index.

- Driver level: `set.seed(42 + sweep_id)` before parameter grid construction.
- Replicate level: `set.seed(20260507)` at the loop top, with `parallel::mc.set.seed` inside `mclapply`.
- The two seeding stages operate at distinct pipeline points and do not collide.

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Seeding operates at two levels. At the driver level, each sweep calls `set.seed(42 + sweep_id)` before constructing its parameter grid, so the parameter grid and any pre-loop draws are reproducible per sweep. At the replicate level, each driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`; per-replicate seeds derive from this master seed by adding the replicate index. These two seedings operate at distinct stages of the pipeline and do not collide.

[54]

All file paths for reproducing this manuscript are fixed: per-replicate RDS, ADEMP summary, driver scripts, pre-registered plan, and manuscript source.

- Per-replicate outputs: `analysis/data/quick-sim/05-design-replicates.rds`;
ADEMP summary: `05-design-summary.txt`.
- Driver scripts: `analysis/scripts/nof1-design-sensitivity/`
and `analysis/scripts/quick-sim/05-design-prototype.R`.
- Pre-registered ADEMP plan: `00-ademp-pre-registration.md`;
manuscript source: `analysis/report/05-nof1-design-sensitivity/`.

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tempor incididunt ut labore et dolore magna aliqua.*

Per-replicate simulation outputs are checkpointed at `analysis/data/quick-sim/05-design-replicates`
Morris ADEMP cell-level summaries at the companion `05-design-summary.txt`.
Driver scripts are under `analysis/scripts/nof1-design-sensitivity/`
and `analysis/scripts/quick-sim/05-design-prototype.R`. The
pre-registered ADEMP plan is at `analysis/scripts/nof1-design-sensitivity/00-ademp-pre-regist`
The manuscript source, paper-specific bibliography (`references.bib`),
and SIM CSL file are at `analysis/report/05-nof1-design-sensitivity/`.

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